

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Odd Semester)

Innovative Practices Description

UNIT I Transmission Line Theory

Degree, Semester & Branch: V Semester B.E ECE A

Course Code & Title: EC6503 Transmission Lines and Wave Guides

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: Types of Transmission Lines

Name of the Innovative Practice: Real Time Images (Demonstration)

Duration: 20 minutes

Description:

A Transmission Line is a device designed to guide electrical energy from one point to another. It is used, for example, to transfer the output RF energy of a transmitter to an antenna. This energy will not travel through normal electrical wire without great losses. In general, Transmission lines are used for purposes such as connecting radio transmitters and receivers with their corresponding antennas, distributing respective cable television signals, trunk lines routing calls between telephone switching centers, computer network connections, and high speed computer data buses.

Types of transmission lines include parallel wire line, twisted pairs, co-axial cable, wave guides and fiber optical wires as shown in figure.



Parallel lines



Co-axial cable



Twisted Pair



Rectangular Wave Guide



Circular Wave Guide



Fiber cable

I have shown all real cables to the students and elaborated its advantages and disadvantages. I explained types of losses associated with all cables. After this demonstration, students also started to collect different types of cables and acquired better knowledge about various transmission lines and had a better learning in the classroom.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3										1

CO1: The student will be able to analyze the various types of transmission lines and the losses associated

References:

1. <https://www.sciencedirect.com/topics/engineering/transmission-line-theory>
2. John D Ryder, "Networks, lines and fields", 2nd Edition, Prentice Hall India, 2010.